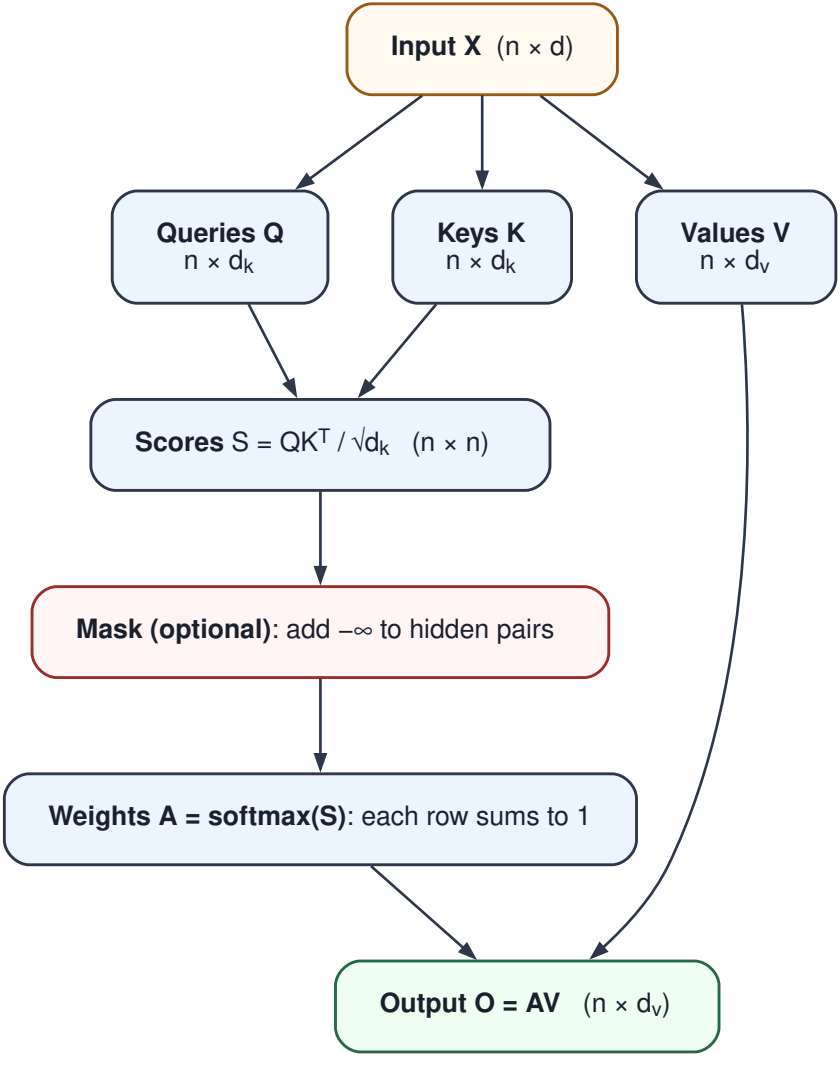
Input X (n × d)
Queries Q
n × d_k
Keys K
n × d_k
Values V
n × d_v
Scores $S = QK^T / \sqrt{d_k}$ (n × n)
Mask (optional): add −∞ to hidden pairs
Weights A = softmax(S): each row sums to 1
Output O = AV (n × d_v)